PHYSICAL SCIENCE
CHEMISTRY TEST

Grade 10 **24 March 2026**

Time: 42 mins **49 marks**

ASSESSOR: K. Bolton **MODERATORS: G. Main and T. Kirsten**

INSTRUCTIONS:

1. This question paper consists of 7 pages, and a data sheet. Please check your question paper is complete.
2. All questions are to be answered on the question paper.

NAME: _____ **SCIENCE TEACHER:** _____

QUESTION 1 MULTIPLE CHOICE

Answer these questions on the multiple-choice grid below by placing a cross over the letter of your choice.

1.1	A	B	C	D
1.2	A	B	C	D
1.3	A	B	C	D

1.1 Element *X* is in group 2, and element *Y* in group 17, of the periodic table. Which ions will be present in the compound formed when *X* and *Y* react together?

A X^{+} and Y^{-}

B X^{2+} and Y^{-}

C X^{+} and Y^{2-}

D X^{2-} and Y^{+}

1.2 Based on electronegativity values, which bond is the most polar?

A B—C

B C—O

C N—O

D O—F

1.3 The number of electrons in the valence shell of elements X and Y are 5 and 7 respectively. What is the formula and type of compound formed from these elements?

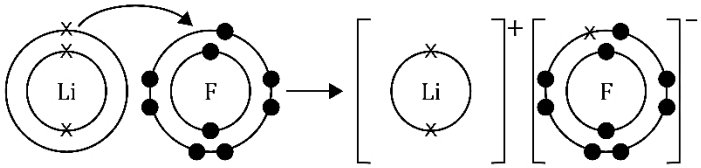
A Covalent, XY_3
B Ionic, XY_3
C Covalent, X_3Y
D Ionic, X_3Y

[2X3=6]

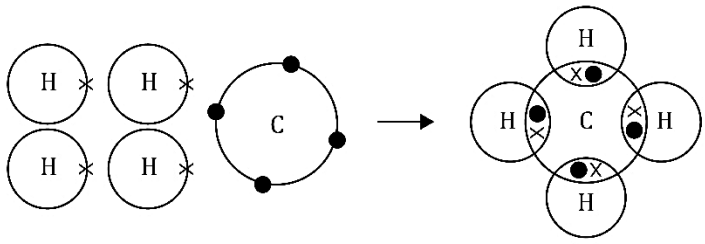
QUESTION 2

The diagrams below illustrate THREE different types of bonding.

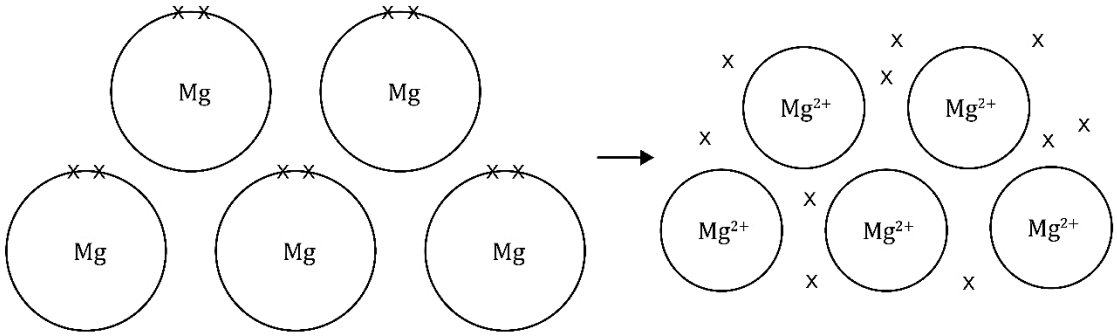
A.



B.



C.



2.1 Complete the sentences comparing the electron distribution in different substances:

(a) In covalent compounds, the bonding electrons are _____.

(b) In ionic compounds, electrons are _____.

(c) In metals, the electrons are _____.

(3)

2.2 In each case, **CIRCLE** the correct word between the brackets:

(a) The type of bonding in figure A is **METALLIC/IONIC/COVALENT.**

(b) The type of bonding in figure B is **METALLIC/IONIC/COVALENT.**

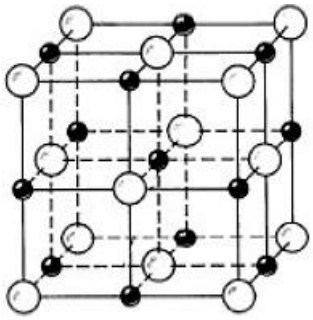
(c) The type of bonding in figure C is **METALLIC/IONIC/COVALENT.** (3)

2.3 Why are metals good conductors of electricity? (2)

_____

_____

2.4 The diagram illustrates the crystal lattice structure of solid sodium iodide.



CIRCLE the correct word within the brackets. (1)

The particles represented by  in the lattice structure are

(ATOMS / MOLECULES / IONS).

2.4.1 Explain whether this substance would conduct electricity when solid OR in solution. (3)

[12]

QUESTION 3

Equal sized pieces of lithium, magnesium and copper are placed in separate test tubes containing hydrochloric acid.

3.1 Which of these metals would be the fastest to react with the acid? (1)

3.2 Which, if any, of these metals would NOT react with the acid? Give a reason for your answer. (2)

3.3 Define the term *electronegativity*. (2)

3.4 Draw a Lewis diagram to show the formation of a bond in a molecule of hydrochloric acid. (3)

3.5 Using electronegativity numbers, state the SPECIFIC type of bonding in hydrochloric acid. (3)

3.6 Magnesium reacts with hydrochloric acid to form a salt and a gas.

3.6.1 Write a balanced chemical equation to show the reaction between magnesium and hydrochloric acid. (4)

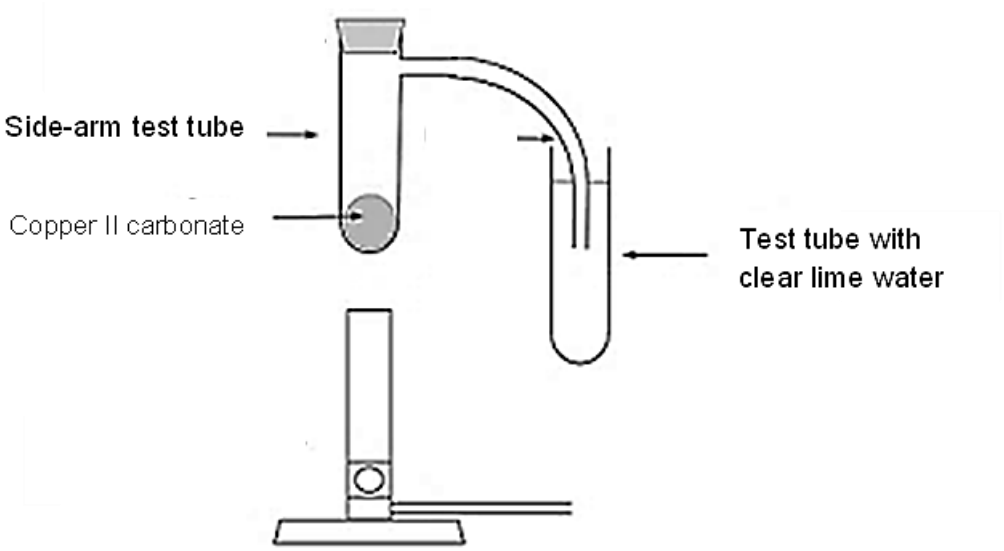
3.6.2 NAME the type of reaction between magnesium and hydrochloric acid? (1)

3.6.3 Describe how you would test for the gaseous product? (2)

[18]

QUESTION 4

Copper (II) carbonate is heated, and the products are a solid and a gas. The gas that is released, is bubbled through clear limewater.



4.1 State the colour change that would be observed in the side arm test-tube. (2)

4.2 NAME the TYPE of reaction. (1)

4.3 NAME the solid product produced from this reaction. (2)

4.4 NAME the gas product and give the result of the limewater test. (2)

4.5 **Lead (II) nitrate + ammonium hydroxide → ammonium nitrate + lead (II) hydroxide**

4.5.1 Rewrite the equation above in its balanced form, using formulae. (5)

4.5.2 NAME the TYPE of reaction. (1)

[13]